# RNA-seq解析パイプライン上級:
# *de novo* assemblyほか

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# *de novo* RNA-seq assembly

Millions of short reads

**pre-processing**

**mapping**

Reads aligned to reference

**summarization by unit**
(gene, transcript, exon)

Table of counts

**normalization**

**DE testing**

List of DE gene

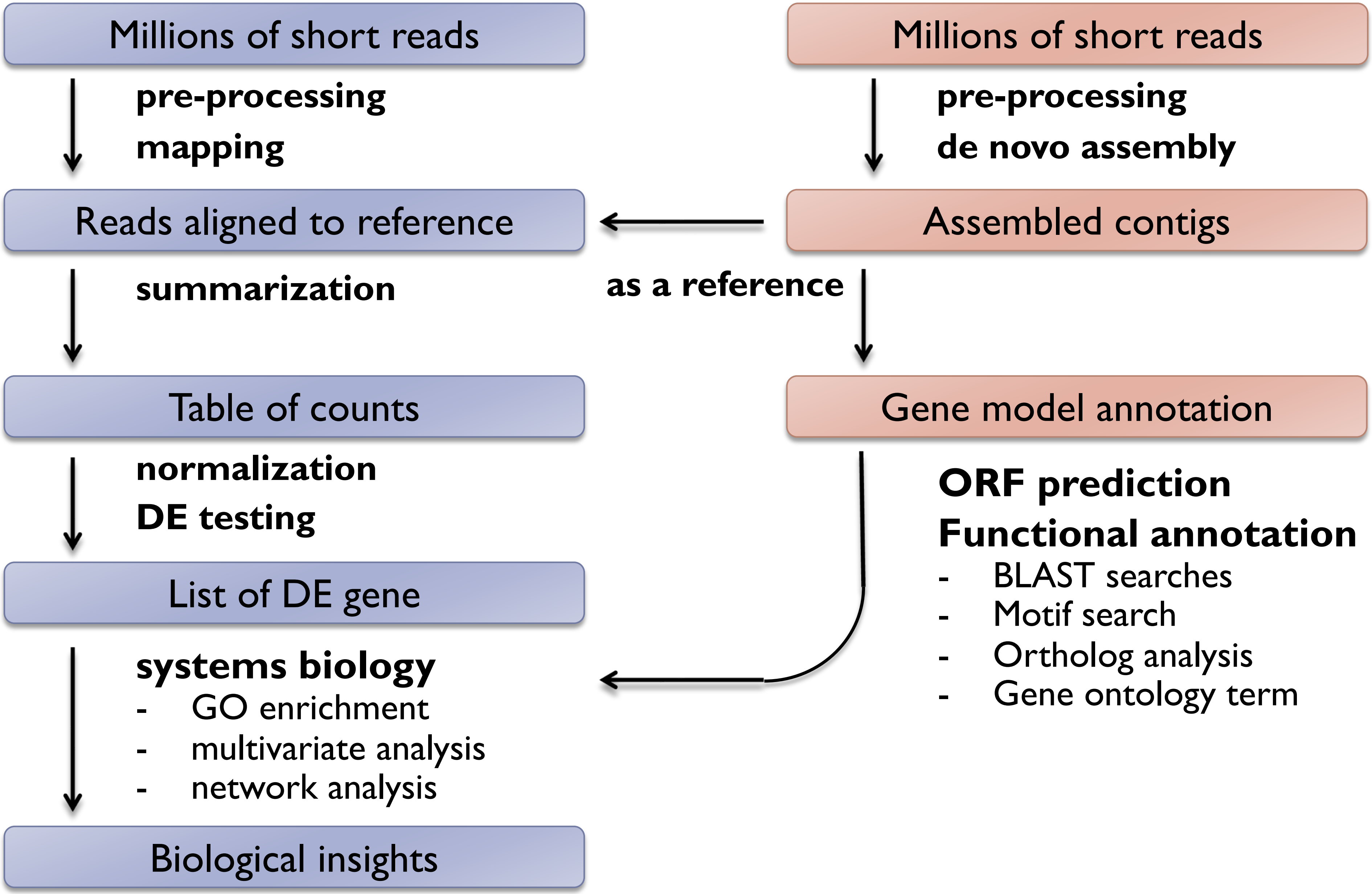
**systems biology**

- GO enrichment
- multivariate analysis
- network analysis

Biological insights

1. **Build** reference
2. **Characterize** reference

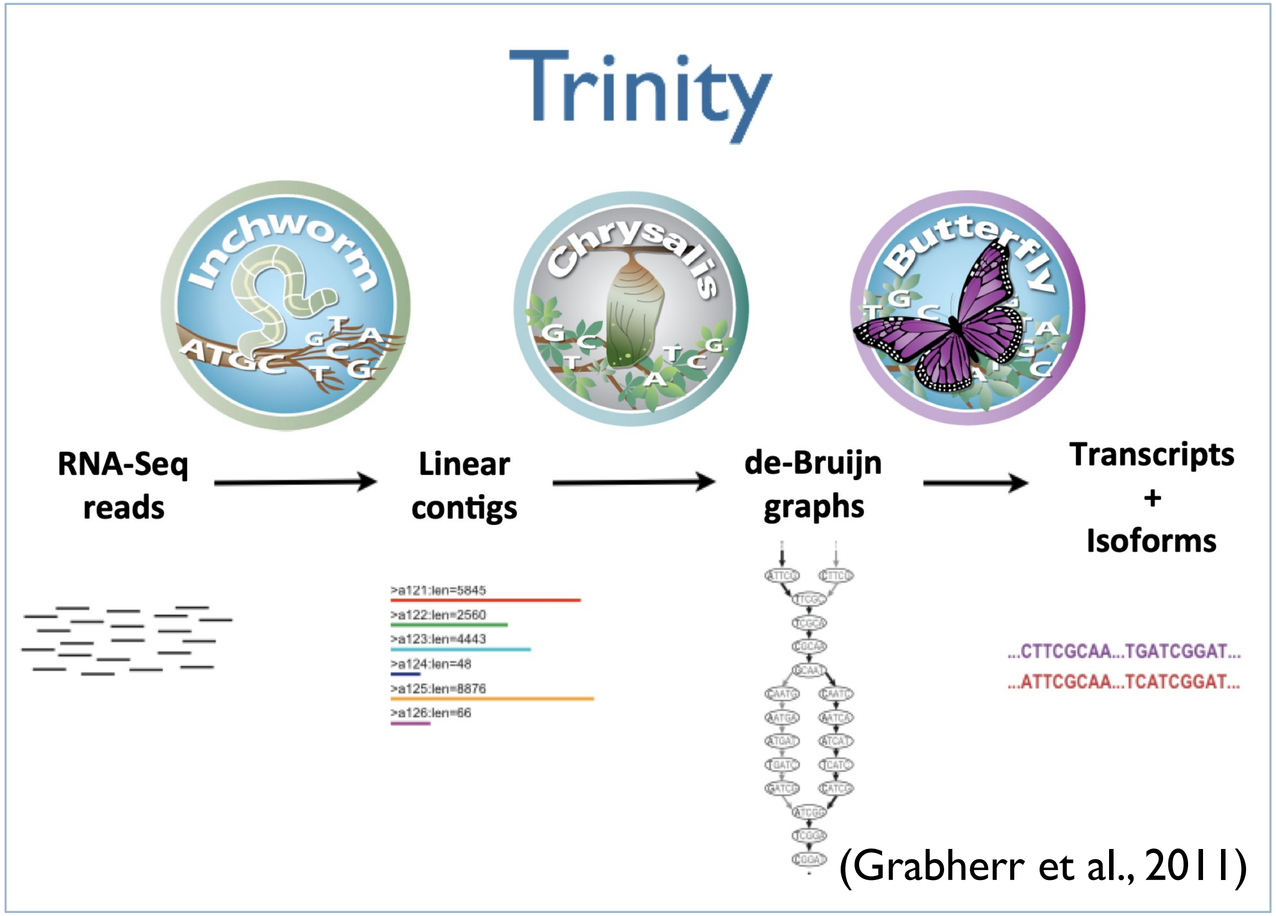
# RNA-seq analysis pipeline (***de novo*** strategy)



# *de novo* assemblers of RNA-seq

*De novo* assemblers use reads to assemble transcripts directly, which does not depend on a reference genome.

- Trinity
- Oases
- TransAbyss
- …




(Grabherr et al., 2011)

https://github.com/trinityrnaseq/trinityrnaseq/wiki trinityrnaseq / **trinityrnaseq**

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Code | Issues 25 | Pull requests 0 | Projects 0 | Wiki | Insights

# Home

https://github.com/trinityrnaseq/trinityrnaseq/wiki

Brian Haas edited this page on Nov 1, 2017 · 35 revisions

# RNA-Seq De novo Assembly Using Trinity

## Quick Guide for the Impatient

Trinity assembles transcript sequences from Illumina RNA-Seq data.

Download Trinity here.

Pages 30

# Trinity example

- Input: Illumina short reads in FASTQ | FASTA format
- Output: assembled contigs in FASTA format

```
# Run Trinity
$ Trinity --seqType fq --left left_all.fq --right right_all.fq \
         --CPU 8 --max_memory 20G
```

(Trinity is supported on only Linux)

# Let's try Trinity assembly

- ex701: *de novo* RNA-seq assembly using Trinity

demo

# Evaluate assembly

- Assembly stats
  - Number of contigs
  - Total length
  - mean, median, N50
- Coverage
  - BUSCO
  - Map back input reads
  - Map other RNAseq reads / known transcripts
- Contamination
  - BLAST (diamond) nr

# Typical assembly stats

```
################################
## Counts of transcripts, etc.
################################
Total trinity 'genes': 7092
Total trinity transcripts: 7269
Percent GC: 44.84

########################################
Stats based on ALL transcript contigs:
########################################
Contig N10: 1255
Contig N20: 914
Contig N30: 714
Contig N40: 564
Contig N50: 453
Median contig length: 301
Average contig: 414.20
Total assembled bases: 3010808
```

# N50

- Definition: Scaffold/contig length at which you have covered 50% of total assembly length
- Larger N50 means longer contiguity

Example and how to calculate

You have 5 contigs:

- 200, 800, 100, 1500, 1000 bp

Calculate total length => 3600 bp

Sort by length and calculate cumulative length

Find the contig at which cumulative / total length reach 50%

| Contig size | Cumulative (bp) | Cumulative / total length |
|---|---|---|
| 1500 | 1500 | 41.6% |
| **1000** | 2500 | 69.4% |
| 800 | 3300 | 91.6% |
| 200 | 3500 | 97.2% |
| 100 | 3600 | 100% |

N50

50%

# N50

- N50の求め方
  - seqkit ソフトウェアなど
  - 定義は簡単なので自分でコードを書く

(R)

```
> len <- c(200, 800, 100, 1500, 1000)
> len.sorted <- rev(sort(len))
> N50 <- len.sorted[cumsum(len.sorted) >= sum(len.sorted)*0.5][1]
```

# BUSCO

https://busco.ezlab.org/

BUSCO

UNIVERSITÉ DE GENÈVE
FACULTÉ DE MÉDECINE

SIB Swiss Institute of Bioinformatics

# BUSCO

from QC to gene prediction and phylogenomics

**BUSCO v5.0.0 is the current stable version!**
Gitlab, a Conda package and Docker container are also available.

Based on evolutionarily-informed expectations of gene content of near-universal single-copy orthologs, BUSCO metric is complementary to technical metrics like N50.

## Availability

- Git source code
- Docker container
- Conda package

## New in v4

- Bacteria & archaea revised
- Auto-lineage selection
- Automated download of datasets

## vs CheckM

- Scores eukaryotes and prokaryotes
- Can run on a laptop
- Better resolution, less overestimates

# BUSCO

BUSCO provides a quantitative assessment of the completeness in terms of expected gene content of a genome assembly or transcriptome by using universally conserved one-copy gene set. The results are simplified into categories of Complete and single-copy, Complete and duplicated, Fragmented, or Missing.

```
# Run BUSCO
$ busco -m transcriptome contigs.fa -o OUTPUT -l lineage
```

```
# example of output
    (Insecta)
    C:94.5%[S:88.5%,D:6.0%],F:1.1%,M:4.4%,n:978

    925 Complete BUSCOs (C)
    866 Complete and single-copy BUSCOs (S)
    59  Complete and duplicated BUSCOs (D)
    11  Fragmented BUSCOs (F)
    42  Missing BUSCOs (M)
    978 Total BUSCO groups searched
```

練習：ex702

Advanced

# Clean up reference sequences

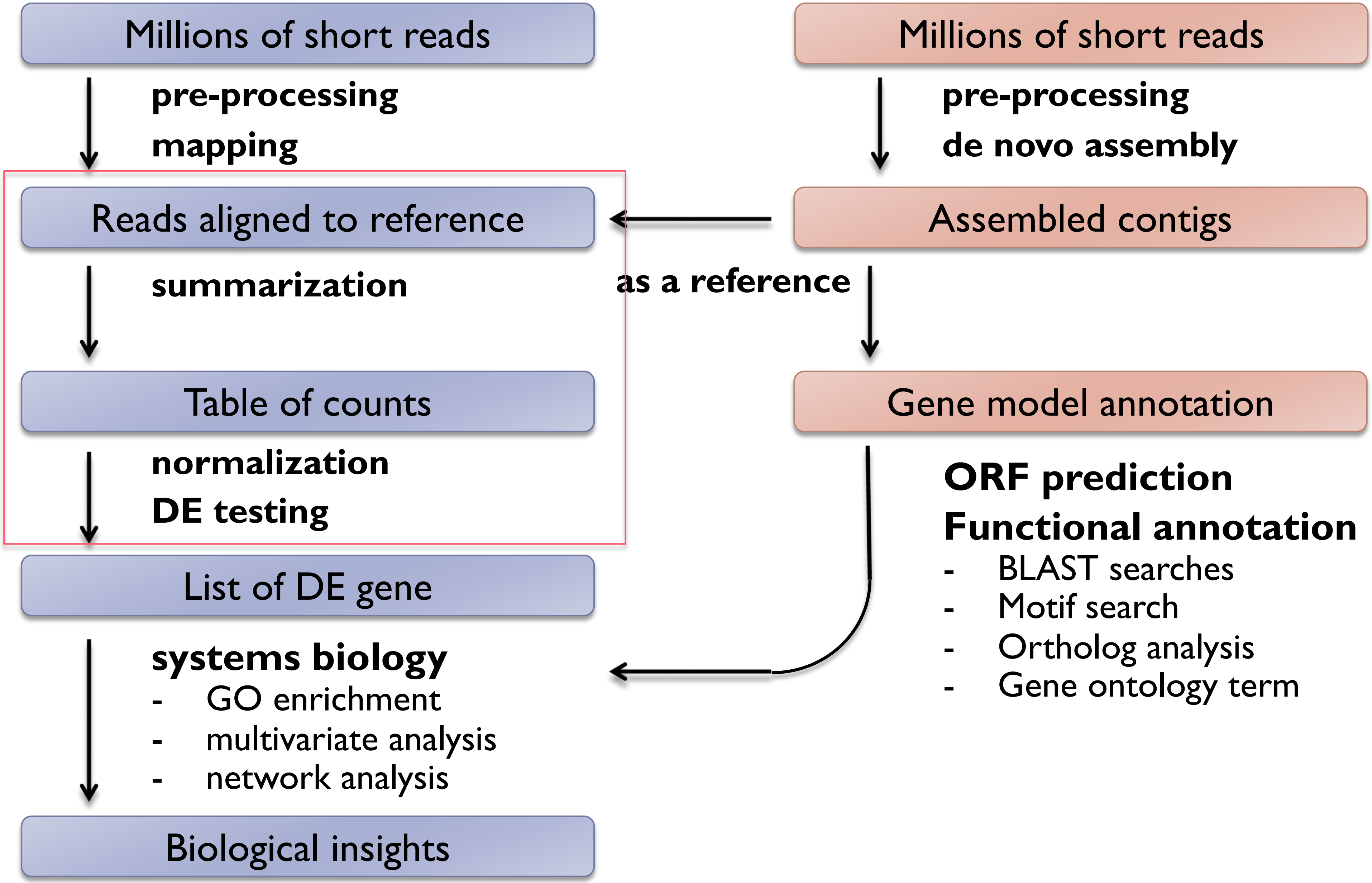
- An issue: Inflation of the number of Trinity contigs is often observed.
  - Trinity outputs splicing variants separately
  - Contaminations
  - Artifacts (bad contigs)
  - Incomplete contigs with very low expression.
- Solution
  - Filter out unwanted contigs.
  - Filter out very lowly expressed transcripts.
  - Cluster similar sequences.

Advanced

# Remove redundancy in reference sequences

- Strategy and Tools
  - Choose one representative transcript from each cluster based on Trinity component information. (longest or highest expression)
  - Clustering
    - CDHIT-EST (http://weizhongli-lab.org/cd-hit/)
    - Corset (Davidson et al., 2014).
    - RapClust (https://github.com/COMBINE-lab/RapClust)
    - EvidentialGene (http://arthropods.eugenes.org/EvidentialGene/trassembly.html)
- Advantage of redundancy reduction
  - Gene-oriented analysis => easier interpretation
  - Better control of multiple comparison.

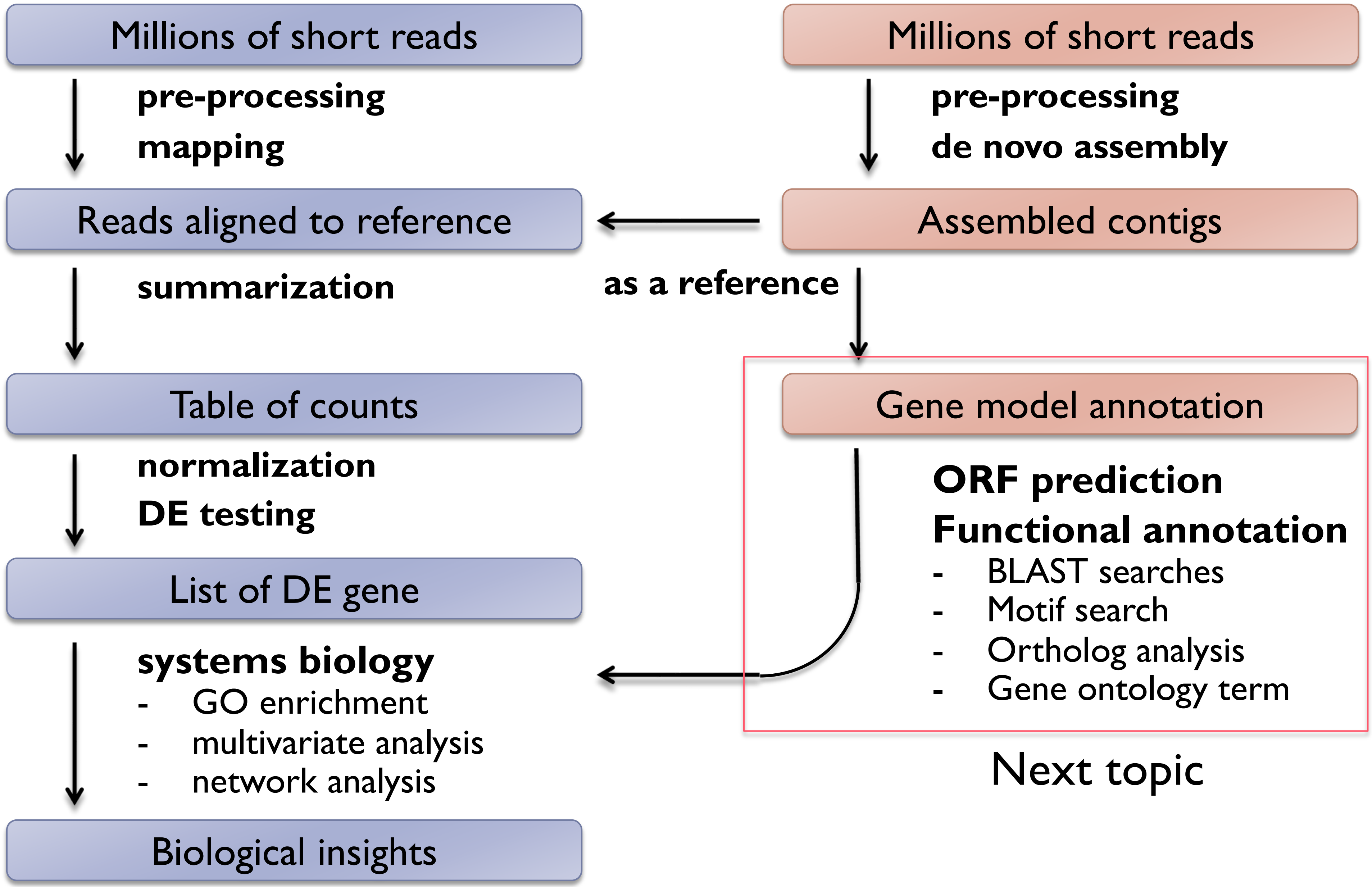
# RNA-seq analysis pipeline (*de novo* strategy)



# DEG analysis

- Follow transcript-based RNA-seq pipeline

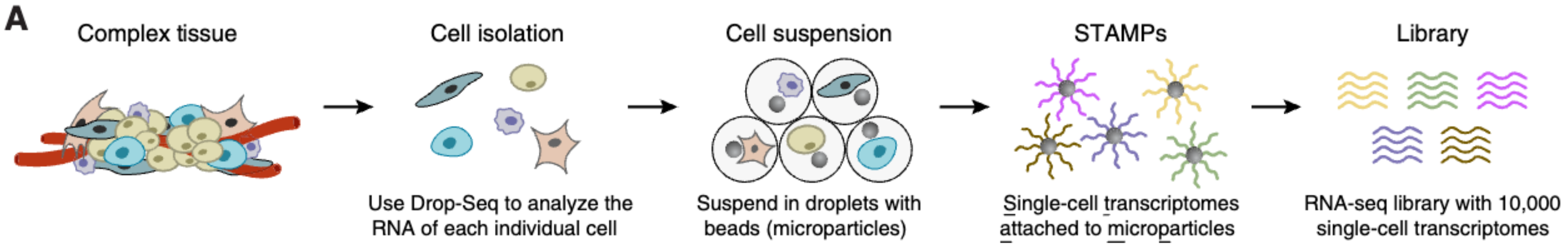
# RNA-seq analysis pipeline (*de novo* strategy)



# Single-cell RNA-seq

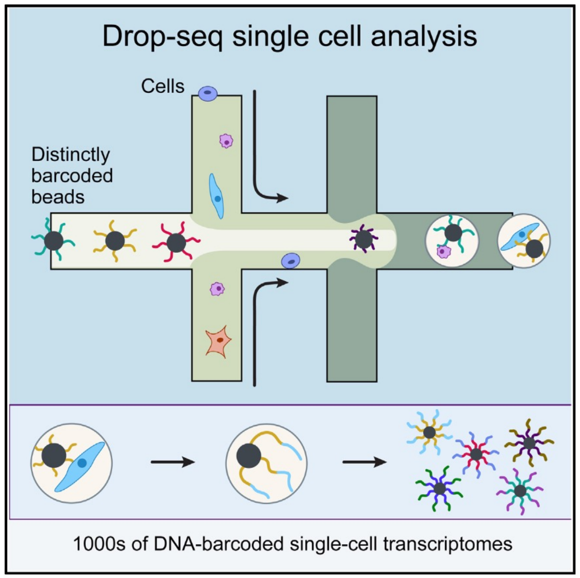
Quick introduction

# Drop-seq / Single-cell RNA-seq




## Highly Parallel Genome-wide Expression Profiling of Individual Cells Using Nanoliter Droplets

### Graphical Abstract





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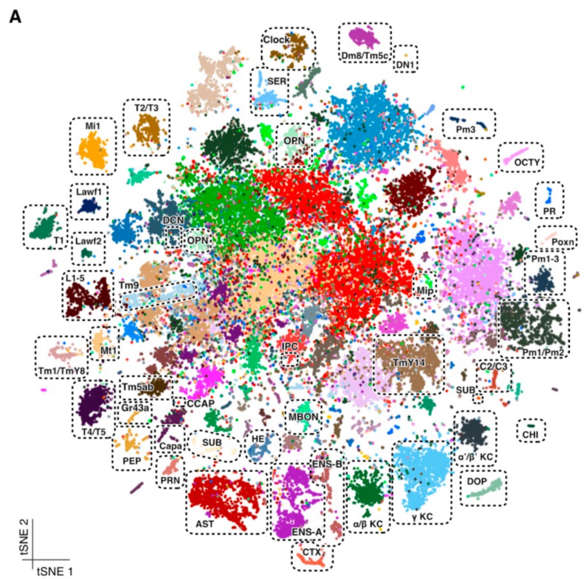
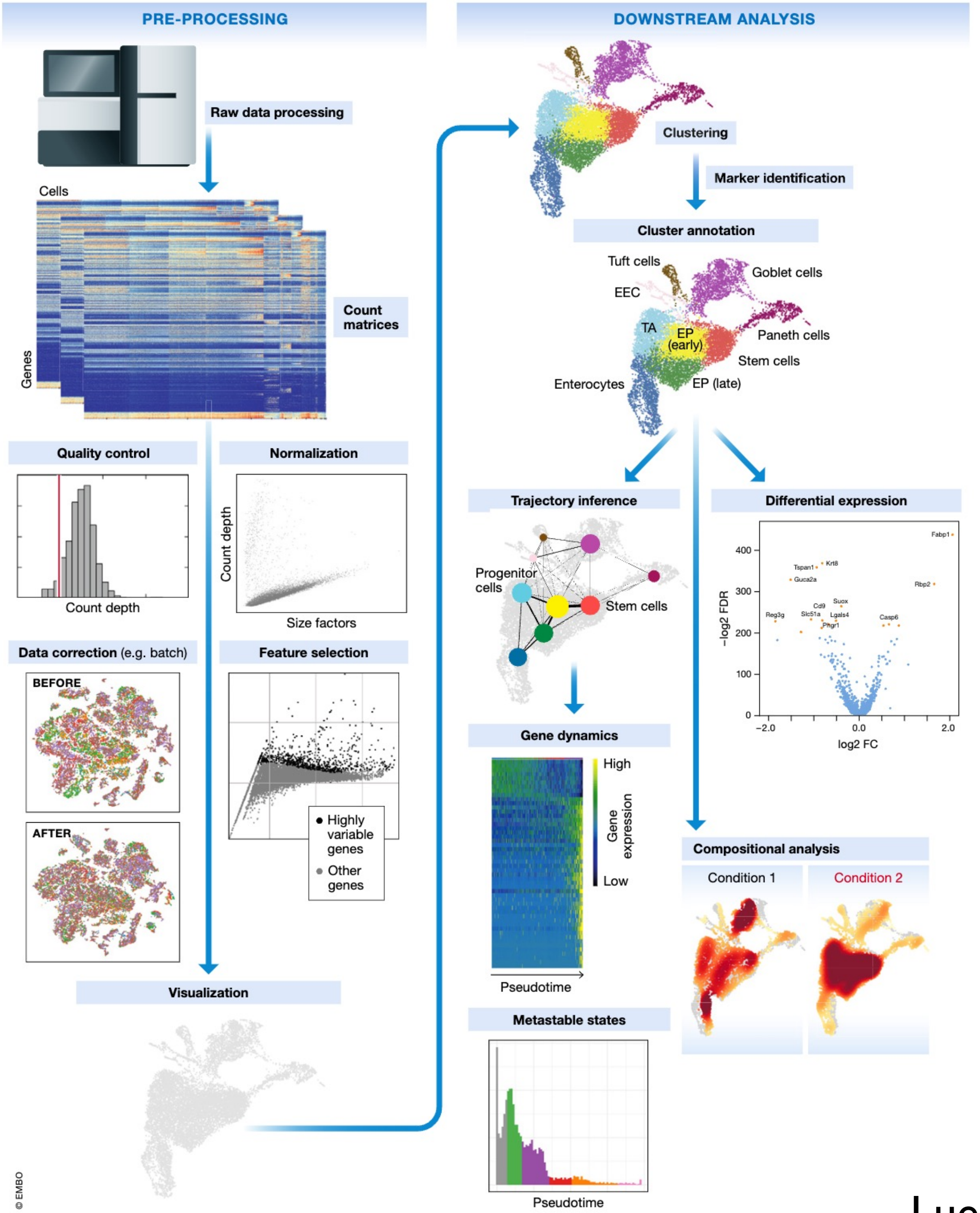
### In Brief

Capturing single cells along with sets of uniquely barcoded primer beads together in tiny droplets enables large-scale, highly parallel single-cell transcriptomics. Applying this analysis to cells in mouse retinal tissue revealed transcriptionally distinct cell populations along with molecular markers of each type.




(Macosko et al., 2015)

# Typical single-cell RNA-seq bioinformatics workflow

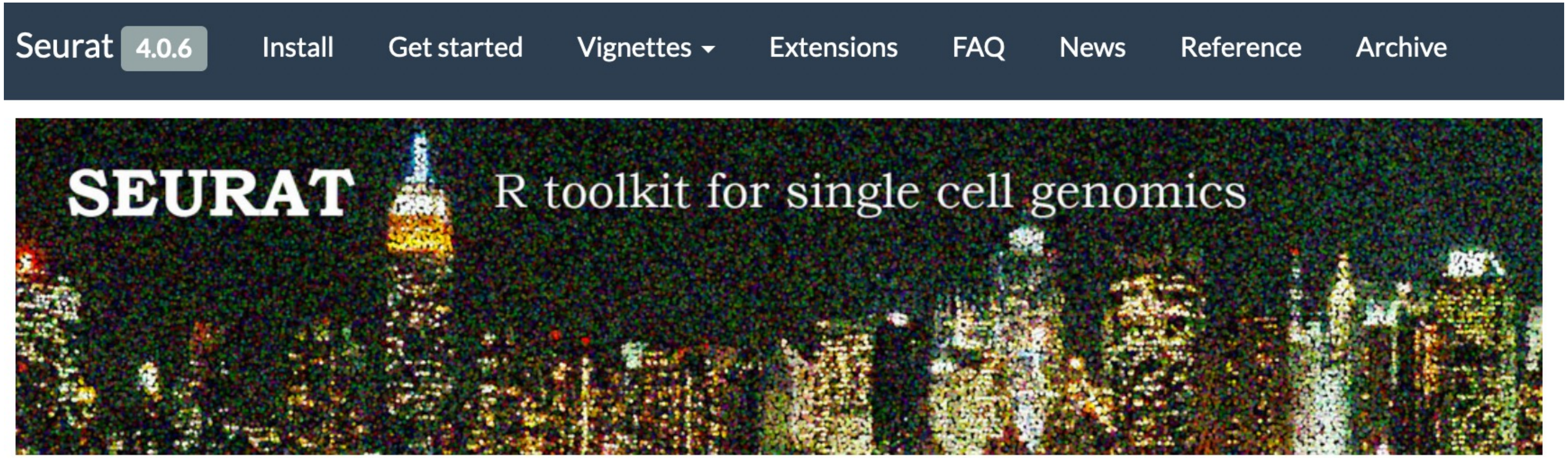



Luecken et al., 2022

# Bioinformatics of single-RNA-seq

- 代表的なプラットフォーム 10x Genomics Chromium. 数千細胞の transcriptome。その他、PIP-seqなど。
- 観測細胞が多いだけで、genes x cells のカウントマトリックスを扱う点は、Bulk RNA-seq と同じ。したがってバイオインフォマティクスの基礎は同じ。
- とはいえ、RNA-seq特有の問題も多く、scRNA-seqに特化したアルゴリズム・ソフトウェアが活発に開発されている。
- Bulk RNA-seq と異なる点
  - Sparce data (ゼロカウントの遺伝子が多い）。それゆえ、データはnoisy。
  - UMIを導入しているプラットフォームでは、生のリードカウントではなくUMIを使う。
  - 観測細胞が桁違いに多い
  - 細胞のクラスタリングに重きを置いた解析が多い
  - scRNA-seqならではの解析。pseudotime 解析など
- Popular tools
  - CellRanger: 10x Genomics社純正　QC + mapping + count matrix generation
  - Seurat: integrated analysis platform (from QC to clustering)
- Geneformerなど、deep learning モデルへの期待

# Toolkits for single-RNA-seq




Seulat https://satijalab.org/seurat/ R

Bioconductor
OPEN SOURCE SOFTWARE FOR BIOINFORMATICS

scanpy

R

Python